

BST Confidence Scale

Two tracks, one discipline. CI data uses D/I/C/S tiers. Human documents use calibrated percentages. Both name the gap.

Track 1: CI Data Layer (D/I/C/S)

Canonical tier system for `data/*.json`, AC theorem graph, and all programmatic-access points.

Tier	Meaning	Criteria
D (Derived)	Mechanism proved, computation verified	Theorem with proof + toy with PASS
I (Identified)	Match <1%, mechanism plausible but not proved	Numerical match + structural argument
C (Conditional)	Depends on named conjecture or unproved lemma	Must name the dependency
S (Structural)	>2% match or qualitative only	Honest about precision limit

Rules: - Every entry in `data/*.json` has exactly one `cal_tier` field from {D, I, C, S}. - No entry has tier “?” — untiered entries must be resolved before EOD. - The older A/B/C/X system is deprecated. Use `cal_tier` as canonical. - Status vocabulary is fixed: {proved, derived, identified, conditional, structural, observed}.

Track 2: Human Documents (Calibrated Percentages)

For OneGeometry, paper abstracts, README current-stats, and narrative documents.

Range	Meaning	What's published	Gap requirement
< 50%	Sketch — early structural reading, no proof attempt	Internal notes only	None
50-80%	Approach identified, key ingredients in hand, gaps unresolved	Working papers, internal	Optional
80-90%	Solid argument but key step missing or unverified	Draft circulated for feedback	Must name the gap
90-97%	Most steps proved, named gaps remain	Pre-submission drafts	Must name each gap
97-99%	Closure complete, may need confirmation or final referee pass	Submitted or near-submission	Must name what the 1-3% is

Range	Meaning	What's published	Gap requirement
Proved	Ready for submission — team + cold reader sign off	Target journal named	Cold reader PASS required
Submitted	Out for review, journal named	Submission record	Journal + date recorded
Under Review	Objections received, list named	Review-process record	Each objection listed
Accepted	Through review	Final	—

Three types of 99%

Type	Meaning	Reviewer response invited
99% (formal-complete)	All steps proved; gap is editorial precision only	“Publish it”
99% (gap-named)	Single proof chain complete except one named subproblem	“Tackle Conjecture X”
99% (want-redundancy)	One route technically sufficient; desire second route	“Find a second route”

The Gap-Naming Rule

Any claim labeled $\geq 80\%$ must name the gap explicitly.

Examples that pass: - “RH 95% — Theorem 6.5 Step 3 spectral-parameter correspondence open. Conjecture 6.1 acknowledged.” - “BSD 99% (gap-named) — rank 0-2 unconditional; rank ≥ 3 conditional on Conjecture 3.2 (R-2 BBW dictionary).” - “ $P \neq NP$ 99% (formal-complete) — three editorial fixes applied (Cal audit May 9). No structural gap.”

Examples that fail: - “~98%.” — No gap named. - “YM ~97%, almost closed.” — “Almost closed” is narrative, not a gap. - “99.5%, ready to submit.” — Doesn’t say what the 0.5% is.

If you can’t name the gap, you don’t know the percentage.

Who Does What

Role	Responsibility
Lyra/Keeper/Elie/Grace Casey + team	Maintain D/I/C/S tiers in data/. Cal-tier is canonical. Assign percentages in human documents. Each $\geq 80\%$ carries its gap statement.
Cal (cold reader) Keeper	Blocking sign-off required before any paper reaches “Proved” status. Maintain review-process objection list. Track submissions, rounds, addressed objections.

Cold-Reader Sign-Off

A paper cannot be labeled “Proved — Ready for Submission” without an independent cold reader’s PASS. The cold-reader role is structurally defined: - Any CI or human not on the paper’s author list can fill it. - The audit criteria are documented (this file + notes/BST_Referee_Methodology.md). - Multiple cold readers may be available; the role is not person-specific.

Review-Process Tracking

Each submission gets a row in the objection tracker (Keeper maintains). Each round of review adds objections. Each resubmission explicitly cites which objections were addressed. Cumulative across submissions — reviewers see their previous concerns engaged.

Cross-Reference

- D/I/C/S tiers map to percentages loosely, not rigidly:
 - D-tier claims are typically $\geq 90\%$
 - I-tier claims are typically 80-95%
 - C-tier claims carry their conditional explicitly
 - S-tier claims are typically $< 80\%$
 - The mapping is approximate. Tiers describe epistemic category; percentages describe completeness of proof chain. A D-tier result can be 92% if the derivation is complete but the paper hasn’t been cold-read yet.
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“If you can’t name the gap, you don’t know the percentage.” — Casey Koons, May 2026